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Project Title	Product and Category Management System
Week	Week 2

WEEK-2 TASK

Task Name

Product and Category Management System

Abstract

The Product and Category Management System is a database application developed using SQL to manage product and category information efficiently. It stores product details, organizes them into categories, and maintains relationships using primary and foreign keys. The system supports CRUD (Create, Read, Update, Delete) operations and generates category-wise reports.

Introduction

The Product and Category Management System is designed to organize product and category information in a structured database. It helps businesses manage products efficiently by storing product details under different categories. Using SQL, users can create tables, insert, update, delete, and retrieve product information quickly, reducing manual errors and improving data management.

Objectives

- Design **Product** and **Category** tables.
- Define **Primary Key** and **Foreign Key** relationships.
- Store product and category details.
- Perform **Insert**, **Update**, **Delete**, and **Select** operations.
- Generate category-wise product reports.
- Improve data organization and accuracy.

Software Requirements

- **Operating System:** Windows 10/11
- **Database:** MySQL Server
- **Tool:** MySQL Workbench
- **Language:** SQL

SQL-CODE

Create Category Table

```
CREATE TABLE Category (  
  
    CategoryID INT PRIMARY KEY,  
  
    CategoryName VARCHAR(50)  
  
);
```

Create Product Table

```
CREATE TABLE Product (  
  
    ProductID INT PRIMARY KEY,  
  
    ProductName VARCHAR(50),  
  
    CategoryID INT,  
  
    Price DECIMAL(10,2),  
  
    FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)  
  
);
```

Insert Data

```
INSERT INTO Category VALUES  
  
(1,'Electronics'),  
  
(2,'Clothing');
```

```
INSERT INTO Product VALUES
```

```
(101,'Laptop',1,55000),
```

```
(102,'T-Shirt',2,800);
```

Update Data

```
UPDATE Product
```

```
SET Price = 60000
```

```
WHERE ProductID = 101;
```

Delete Data

```
DELETE FROM Product
```

```
WHERE ProductID = 102;
```

Display Report

```
SELECT ProductName, CategoryName, Price
```

```
FROM Product
```

```
JOIN Category
```

```
ON Product.CategoryID = Category.CategoryID;
```

Result

The Product and Category Management System was successfully implemented using SQL. The Category and Product tables were created successfully, data was inserted, updated, and deleted without errors. The SQL JOIN query generated a report displaying product names, categories, and prices accurately.

Conclusion

The Product and Category Management System demonstrates the effective use of SQL for database management. It successfully implements table creation, primary and foreign key relationships, CRUD operations, and report generation. The system provides an efficient and organized way to manage product and category information, making it suitable for inventory and retail management applications.